

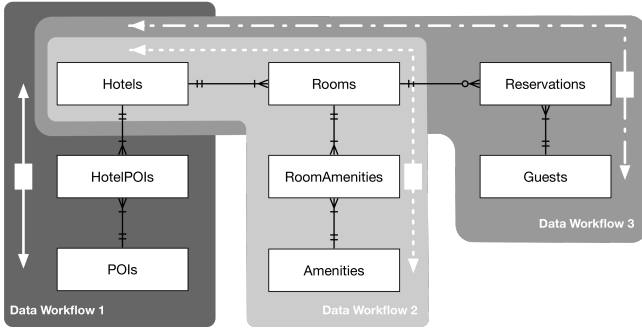


Fig. 7. Data Workflows for the Hotel's Demonstration Case.

Having identified the column-families and the several tables, it is now necessary to assign to each table its respective column-families (Rule CDM.3.1). Each descriptive table integrates the core column-family, or a set of column-families including the core and the complementary descriptive column-families, from which it was derived. Taking $Rooms_T$ as an example, as this table was derived from the complementary descriptive column-family $Rooms_{CF}$, besides this column-family, the HBase table also needs to include the column-family associated with the $Hotels_{CF}$, which is the core descriptive column-family. $Hotels_T$, $POIs_T$, $Guests_T$ and $Amenities_T$ include the corresponding core column-families, namely $Hotels_{CF}$, $POIs_{CF}$, $Guests_{CF}$ and $Amenities_{CF}$.

For the analytical tables (Rule CDM.3.2), $Reservations_T$ integrates four column-families, $Hotels_{CF}$, $Rooms_{CF}$, $Reservations_{CF}$ and $Guests_{CF}$; $RoomAmenities_T$ integrates three column-families, $Hotels_{CF}$, $Rooms_{CF}$ and $Amenities_{CF}$; and, $HotelPOIs_T$ integrates two column-families, $Hotels_{CF}$ and $POIs_{CF}$.

After following the proposed rules, the identified columnar data model for HBase includes five descriptive tables and three analytical data tables, as depicted in Fig. 8 showing the proposed keys (Rule CDM.4) and columns.

The obtained data model allows data management for all descriptive tables in terms of inserting/updating/deleting hotels, rooms, guests, amenities and POIs, as well as upgrading new facts generated by the organization's day-by-day activities, namely new reservations, new amenities for the available rooms or new POIs characterizing the hotels' surroundings. Besides adding these new facts, it is possible to query the available data, answering questions that the operational or tactical managers may have, supporting their decision-making processes.

Looking to the obtained model and comparing it with the one presented in [45], it is possible to verify that although both models present a similar number of tables (seven vs. eight), they are organized in a different way. While the model presented here includes descriptive tables for inserting new POIs or amenities, the model proposed in [45] only includes base tables for hotels, guests and rooms. Reservations and POIs by hotel are organized in a similar way in both models.

The main difference is associated with the possibility of managing all the available data using the approach proposed here, against the possibility of having some specific tables for querying the data, like the available rooms or hotels by city, using the model proposed in [45]. Nevertheless, both data models are able to answer the same questions, as they include the same base columns.

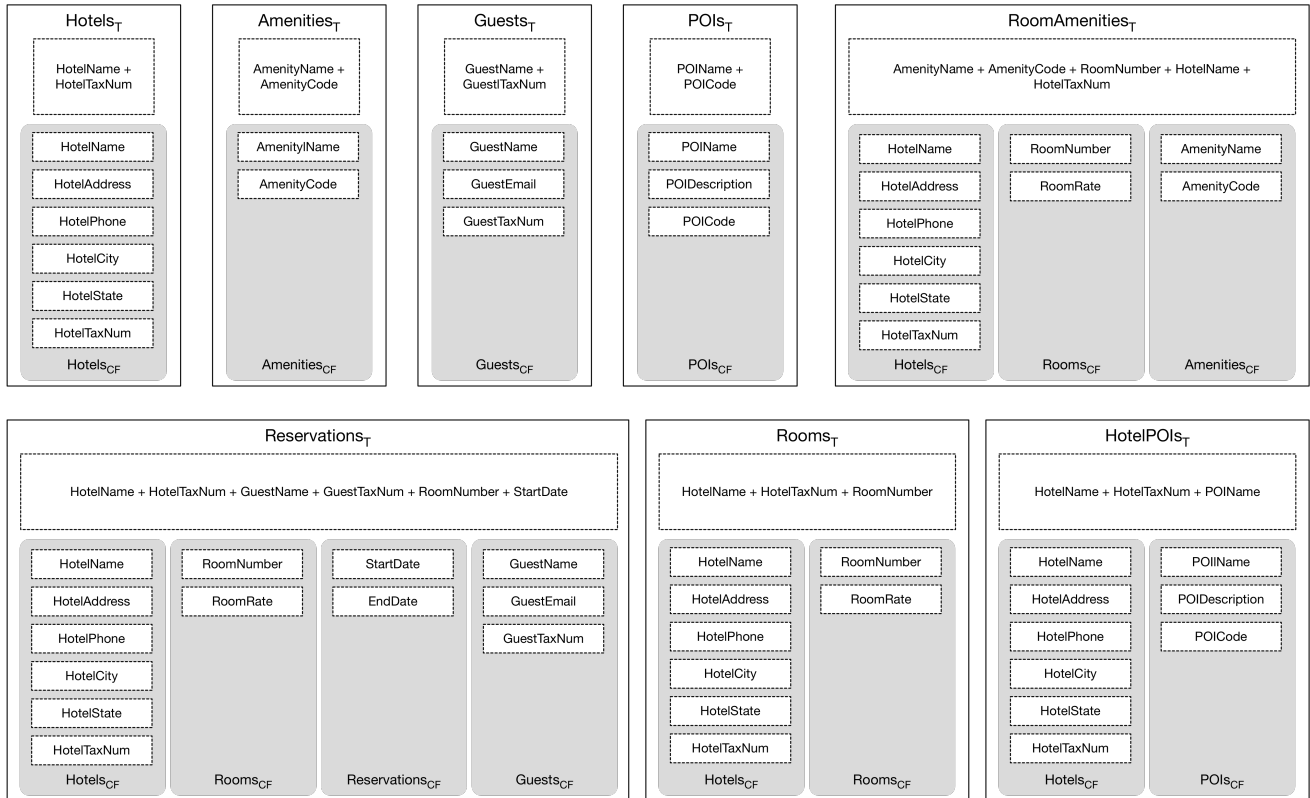


Fig. 8. CDM for the Hotel's Demonstration Case.

At this point, it is important to mention that while the data model here obtained (Fig. 8) is suited for HBase, the one proposed in [45] is suited for Cassandra, highlighting the many different ways of organizing a columnar data schema.

After showing how the proposed rules can be applied, a similar exercise is now followed for transforming a well-known *RDM* into a *CDM*. The TPC Benchmark™ H (TPC-H) is a decision support benchmark that includes a set of business oriented ad-hoc queries and concurrent data modifications [46]. It is usually used for benchmarking decision support systems that process large volumes of data, justifying the transformation of its data model into a *CDM*. The obtained *CDM* can afterwards be used for benchmarking processing technologies with data stored in denormalized tables.

The data model of the TPC-H benchmark is depicted in Fig. 9. As can be seen, this data model includes eight transactional tables dealing with sales, customers and suppliers, modelling a business that manages, sells and distributes products.

Following the proposed rules, and starting by Rule CDM.1, Rule CDM.1.1, two types of descriptive column-families are identified. The core descriptive column-families are *Region_{CF}* and *Part_{CF}*, as these tables only receive relationships with cardinality of 1, not integrating any FK. Three complementary descriptive column-families are identified, giving origin to *Nation_{CF}*, *Supplier_{CF}* and *Customer_{CF}*. Three other tables give origin to analytical column-families (Rule CDM.1.2), namely *PartSupp_{CF}*, *Orders_{CF}* and *LineItem_{CF}*. The attributes of the identified column-families are the attributes of the corresponding tables in the *RDM*, excluding the keys, either PK or FK.

Regarding Rule CDM.2, the descriptive tables are *Region_T*, *Part_T*, *Nation_T*, *Supplier_T* and *Customer_T* associated with the identified descriptive column-families (Rule CDM.2.1), while for the identification of the analytical tables there is the need of identifying the data workflows of the *RDM* (Rule CDM.2.2).

Using all the tables present in the *RDM* that were not identified as descriptive column-families, *PartSupp*, *Orders* and *LineItem*, three data workflows are identified (Fig. 10) using the *n:1* relationships starting in these tables and following this type of relationship until no more *n:1* relationships are found. Following this procedure, the three data workflows originate three analytical tables, named as *PartSupp_T*, *Orders_T* and *LineItem_T*.

Having identified the column-families and the several tables, it is now necessary to assign to each table its respective column-families. Each descriptive table (Rule CDM.3.1) integrates a core column-family, or a set of column-families including the core and the complementary descriptive column-families, from which it was derived.

Taking *Customer_T* as an example, and as this table was derived from the complementary descriptive column-family *Customer_{CF}*, besides this column-family, the HBase table also needs to include the column-families associated with it, which are the *Nation_{CF}* complementary descriptive column-family and the *Region_{CF}* core descriptive column-family. *Region_T* and *Part_T* include the corresponding core column-families, *Region_{CF}* and *Part_{CF}*, respectively. *Nation_T* includes the core *Region_{CF}* and the complementary *Nation_{CF}*. *Supplier_T* includes *Nation_{CF}* and *Supplier_{CF}* as complementary column-families and *Region_{CF}* as core column-family.

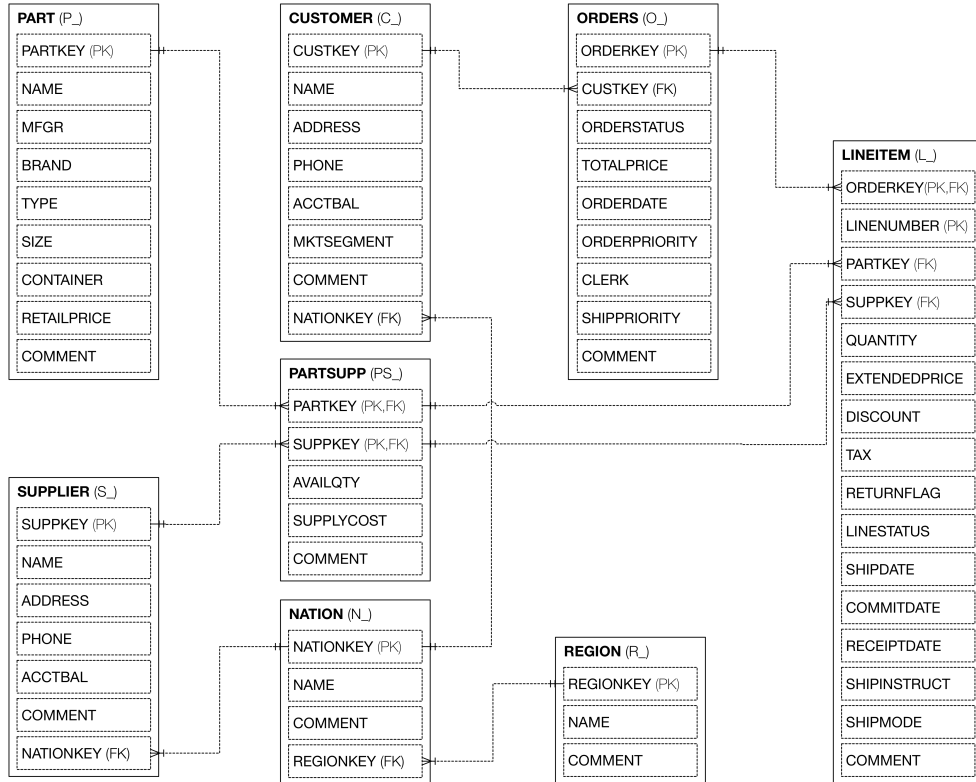


Fig. 9. RDM for the TPC-H data model. Redrawn based on [46].

For the analytical tables (Rule CDM.3.2), *PartSupp_T* includes five column-families, *PartSupp_{CF}*, *Part_{CF}*, *Supplier_{CF}*, *Nation_{CF}* and *Region_{CF}*; *Orders_T* integrates four column-families, *Orders_{CF}*, *Customer_{CF}*, *Nation_{CF}* and *Region_{CF}*; and, *LineItem_T* integrates the eight column-families, achieving a complete denormalization of the relational schema: *LineItem_{CF}*, *PartSupp_{CF}*, *Part_{CF}*, *Supplier_{CF}*, *Orders_{CF}*, *Customer_{CF}*, *Nation_{CF}* and *Region_{CF}*. Moreover, and for this table, it is worth mentioning that *Nation_{CF}* and *Region_{CF}* are denormalized twice, as there is the need of specifying the nation and

region of the supplier, and the nation and region of the customer.

After following the proposed rules, the identified CDM for HBase includes five descriptive tables, as depicted in Fig. 11 showing the proposed keys (Rule CDM.4) and columns. For the analytical tables, and also showing the proposed keys (Rule CDM.4) and columns, Fig. 12 highlights the three obtained analytical tables and the virtual aggregation of column-families to verify the several denormalizations of *Nation_{CF}* and *Region_{CF}*.

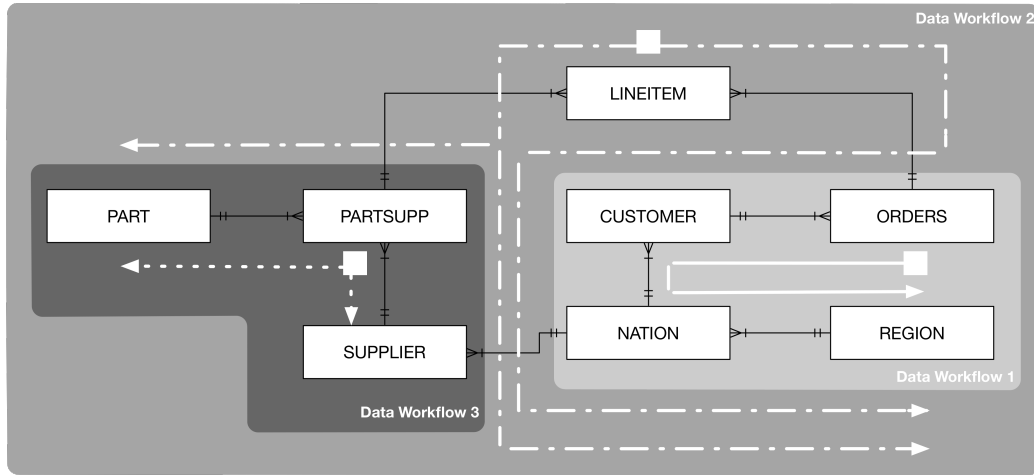


Fig. 10. Data workflows for the TPC-H data model.

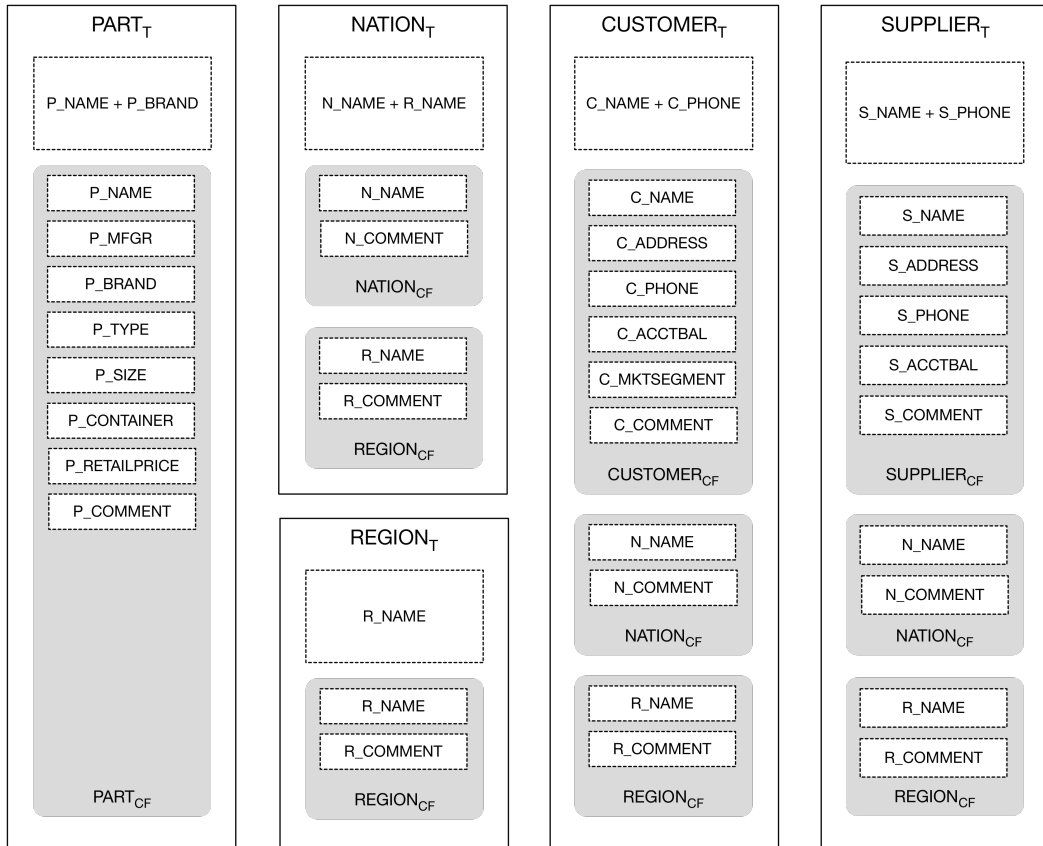


Fig. 11. CDM with the descriptive tables for the TPC-H data model.

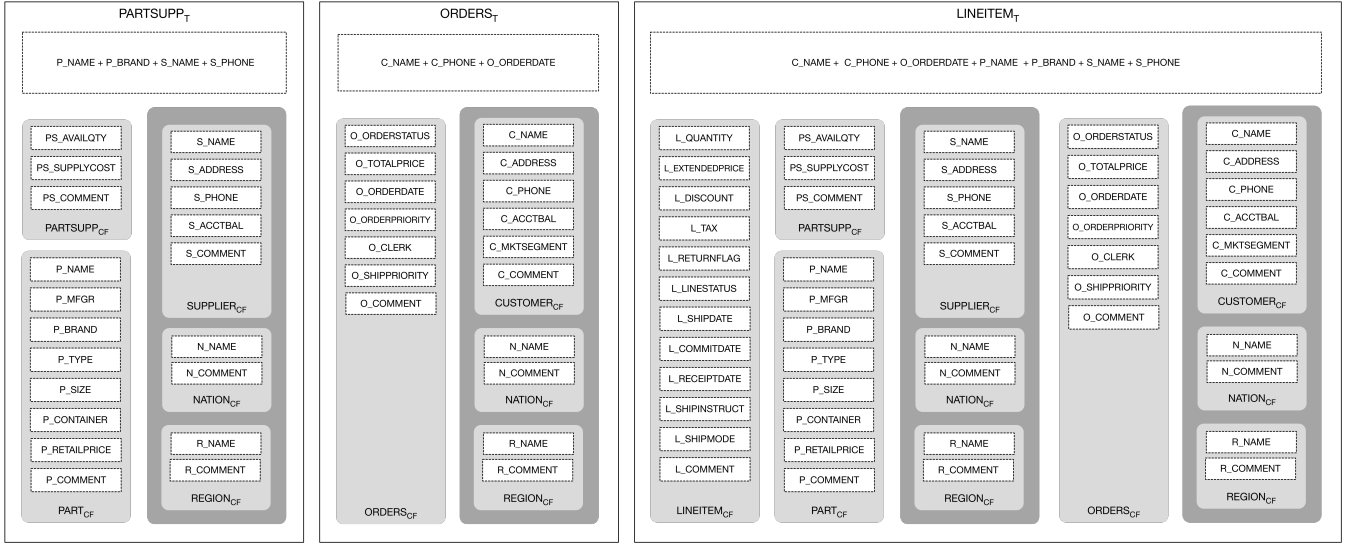


Fig. 12. CDM with the analytical tables for the TPC-H data model.

After following the proposed rules for transforming a *RDM* into a *CDM*, several tables are obtained, addressing different purposes in terms of data management. Some tables have a more operational role, maintaining information about the main entities of an application domain, and some others present an analytical profile, being able to answer specific questions about the business processes of an organization. It is up to the information system engineers to decide which tables must be implemented, having into consideration the data management and analytical tasks that must be supported. In particular, for the analytical tables, *PartSupp_T* and *Orders_T* are fully contained in *LineItem_T*, so all queries can be answered by this last one, being redundant the implementation of the other two tables.

Another issue that depends of the business context in which these tables are going to be used is the definition of the tables' keys. The keys here presented are merely suggestions of how they can be composed and defined for each table. For instance, these can be more appropriate for inserting or searching data, depending on the attributes and concatenation order between them.

V. BIG DATA CHALLENGES

This section presents several challenges regarding Big Data, including general dilemmas, challenges in the Big Data life cycle, issues in security, privacy and monitoring, and required changes in organizations. These challenges are also useful for identifying relevant research topics in this field.

A. Big Data General Dilemmas

General dilemmas may include challenges such as the lack of consensus and rigor in Big Data's definition, models and architectures, for example. In [6], the authors claim that the concept of Big Data is often more commercial speculation than it is a scientific research topic. The authors also mention the lack of standardization in Big Data, such as data quality evaluation and benchmarking. In fact, the lack of standard benchmarks to compare different technologies is seriously aggravated by the constant technological evolution in Big Data environments [47].

Even the ways to fully use Big Data remain an open subject to explore, such as applications in science, engineering, medicine, finance, education, government, retail, transportation or telecommunications, for example [6]. Discussions such as how to select the most appropriate data within several sources, or how to estimate their value, remain as Big Data dilemmas [19]. Another commonly discussed pitfall is how Big Data helps representing the population better than a small dataset [22]. This obviously varies with the context, but the authors call our attention for not assuming that more data is always better.

B. Challenges in the Big Data Life Cycle

These challenges are related to technical difficulties in tasks such as Big Data collection, integration, cleansing, transformation, storage, processing, analysis or governance:

- The need to rethink storage devices, architectures, mechanisms and networks, in order to achieve more efficient input/output (I/O), data accessibility and data transmission [10];
- Scalability becomes crucial to store and analyze data. Handling increasing amounts of data requires redesigning databases and algorithms to extract value from it [27]. Parallel computing becomes crucial to deal with Big Data, assuring availability, cost efficiency and elasticity [6];
- Assuring data quality and adding value through data preparation becomes challenging in Big Data environments [10]. Different data sources may have different data quality problems [27]. These problems and vast amounts of redundancy can also make data integration more difficult [6]. The heterogeneity resulting from multiple sources augment these challenges, as traditional techniques for data analysis expect homogeneous data [48]. Heterogeneity brings implications on data integration [28] and consequences in the analysis of Big Data, since the unstructured nature of data sources presents several challenges regarding transformations to support adequate analytical tasks;

- Visualizing Big Data requires rethinking traditional approaches due to the volume of data, thus joining appearance and functionality is crucial [10]. Advanced data visualizations are needed to extract value from Big Data [49], having the capability to scale to thousands or millions of data points, handle multiple data types, and be easy to use in order to satisfy several users. Manipulating Big Data is challenging due to its characteristics, namely executing drilldowns or rollups. In these visualizations, data from multiple sources is typically integrated into a single picture. Technological evolutions are being made to address the challenge of manipulating Big Data interactively [14];
- Searching, mining and analyzing Big Data is a challenging and relevant research trend, including Big Data searching algorithms, recommendation systems, real-time Big Data mining, image mining, text mining, among others [6]. As [15] claim, size is frequently the main concern in Big Data, but the unstructured nature of data also deserves attention (e.g., text, audio, video) and imposes significant challenges in these tasks;
- Big Data governance faces challenges regarding control and authority over massive amounts of data from different sources [27]. Managing such heterogeneous environment to plan access policies and assure traceability can quickly become almost impossible without adequate governance tools.

Organizations face several challenges in Big Data life cycle. New business problems require technological innovations in the way data flows across the organization. Surpassing these challenges will depend on the organization's maturity, since legacy applications and the use of incompatible formats can impose several difficulties to an adequate integration and extraction of value from Big Data. Collecting data, namely gaining access to it, may also be a challenge, since integrating data from multiple sources, including external ones, raises questions about others' intention to share it free of charge [5].

In [6], the authors state that the efficiency in data flows is key to assure an adequate Big Data processing. The authors also highlight the challenge of building effective computing models in real-time and online applications to analyze Big Data. Other challenges related to processing Big Data may include reutilization and reorganization of data, which become laborious at scale. The characteristics of Big Data require a paradigm shift in databases and analytical technologies, since dealing with Big Data throughout its life cycle can potentially create severe bottlenecks in networks, storage devices and relational databases. Technology is evolving to execute these stages in distributed environments, becoming dependent on high storage capacity and processing power.

Even relational databases are evolving to accommodate these trends, increasing query performance and being able to deal with more data variety [3]. Combining the benefits of RDBMSs and NoSQL databases actually represents a research trend, as well as query optimization in Big Data technologies [28]. Furthermore, advancements are being

made in scalable storage and algorithms. The work of [50] argues that processing queries in Big Data may take significant time, since it is challenging to sequentially iterate through the whole dataset in a short amount of time. Consequently, the authors highlight the relevance of designing indexes and consider adequate preprocessing technologies. In [27], it is identified the need to study adequate models to store and retrieve data, a crucial factor to successfully implement Big Data solutions. Models and algorithms for scalable data analysis also remain an open research issue, as well as the integration and analysis of data arriving continuously from streams. Mining data streams has been identified as an emergent research topic in Big Data analytics [51].

C. Big Data in Secure, Private and Monitored Environments

Nowadays, keeping data secure and private is one of the most concerning tasks for organizations [6], [48]. Users want to rest assure that any leaks into the public domain will not occur [19]. In [52], citing a survey from [53], it is claimed that security and privacy are frequently mentioned among the Big Data concerns of Information Technology (IT) managers. It is relevant to plan a Big Data driven security model for organizations to accurately specify risks and prevent illegal activity or cyber threats. Several considerations are mentioned, such as authentication, authorization, network traffic analysis, data protection laws and mining data related to security. In [6], it is also discussed the potential for Big Data applications related to security concerns.

Due to the characteristics of Big Data, more risks arise and traditional data protection methods must be rethought. The work of [6] argues that Big Data applications face multiple challenges related to security, privacy and monitoring: protection of personal privacy during not only data collection, but also in its subsequent storage and flows; Big Data quality and its influence to appropriate and secure uses; the performance of security mechanisms like encryption is largely influenced by the scale and variety of data; and other aspects related to secure communications, administration and monitoring in environments with multiple users and services. Other relevant challenge, as highlighted by [27], is assuring Big Data integrity, i.e., data is only modified by the owner or other authorized entities.

Policies related to data are also relevant today, at a time when there is a significant amount of sensitive data about individuals, such as the one related with their health or finances [5]. Legal issues are being raised regarding the easiness to copy, integrate and recurrently use data by different people. Intellectual property, data ownership and responsibility regarding inaccurate data, deserve proper attention from policy makers [5]. Legal and regulatory issues deserve attention in several aspects [50], like analyzing the adequacy of current laws and regulations to properly protect data about individuals [27]. Even the constant tracking on employees within an organization can raise discussions about adequate work policies [54].

Besides these issues, the work of [55] raises questions about the implications of having data widely and transparently available. Assuring privacy is both a technical

and sociological problem, as [48] argue. Inadequate availability of location-based data allows the possibility to infer a person's residence, office location and identity, for example. Moreover, many other data sources can contain personal identifiers, or even if no personal identifiers exist, when data is rich enough, one can draw reasonable inferences from it [56]. Another relevant topic, briefly mentioned above, is data ownership, due to its value for certain organizations that are currently debating about ways of sharing or selling data without losing control of it [48]. Data ownership is often discussed regarding social media websites, since users' data is not owned by organizations, although they store it [19]. As [56] discuss, these organizations tend to assume that they hold the rights of the data, and sometimes the current legislation benefits them, allowing organizations to not permanently delete data, even when users ask for it.

Discussing Big Data security, privacy and monitoring in cloud environments is also relevant. Organizations frequently recognize that using Big Data technologies in cloud environments helps reducing their IT costs [50], although raising concerns about Big Data storage and processing infrastructures. Therefore, as [50] claim, one of the challenges lies in assuring adequate monitoring and security without exposing users' data when processing it.

D. Organizational Change

Surely Big Data may sound appealing to most organizations, but frequently organizational leaders lack the understanding of its value and how to extract it [5]. Occasionally, the lack of knowledge in how to use analytics is mentioned as the leading obstacle to become more data-driven [57]. Within several business areas, organizations need to monitor trends and gain advantages compared to their competitors, but as [5] discuss, many of them lack the talent, the rigorous workflows structure and the incentives for adequate Big Data initiatives to better support decision-making. Leaders and policy makers must understand how Big Data can create value, as well as critically thinking about IT capabilities, data strategies, analytical talent and data-driven approaches.

This paradigm shift in organizations requires them to move analytics into the core business and operational functions [3], changing business processes, delivering insights related to customers, products, services, and other transactions. [58] present challenges that organizations will face in management, caused by Big Data initiatives, among which can be highlighted: the need for adequate leadership and data scientists [59], computer scientists or other professionals to deal with Big Data; the adequate understanding and use of Big Data technology; and the need to change organizational culture. Big Data initiatives require a multidisciplinary approach, demanding collaboration to deliver useful results that must be properly understandable by the organization [48], although to accomplish this, significant organizational changes must occur.

VI. CONCLUSION

This paper presented a state-of-the-art literature review in Big Data and also a modelling approach to convert relational databases into column-oriented databases. Besides this

modelling approach, this work discussed the relevance, definition and challenges in Big Data, as well as how to design Big Data solutions through the use of current techniques and technologies.

Big Data is a concept of major relevance in today's world, and its popularity has increased considerably during the last years. Business areas like healthcare, retail, manufacturing and modern cities will benefit from the collection, storage, processing and analysis of Big Data, leveraging unprecedented data-driven workflows and considerably improving decision-making. This new type of data is being defined not only by its characteristics (e.g., volume, variety, velocity), but also by the limitations it imposes in traditional technologies.

Several requirements and techniques to design Big Data solutions were discussed. Moreover, Hadoop and related projects were presented, as well as scalable databases and other technologies for Big Data Analytics. One can conclude that there is no common approach to work with Big Data and there are innumerable technologies to choose from, each trying to stand out, which creates barriers in the design and implementation of Big Data solutions, since most of the time they are competitors, or their role is misunderstood, overlapping each other. Logical architectures discussed in this work solve part of the problem by orchestrating components according to their role and interactions, but ambiguity regarding the Big Data technology most suitable to a specific role still prevails, mainly due to the diverse set of possible choices.

Organizations implementing Big Data initiatives will face many challenges, among which can be highlighted: the lack of consensus in definitions, models or architectures; challenges regarding the Big Data life cycle; concerns related to security, privacy and monitoring; and organizational change, such as new skills and changes in business processes. The proposed modelling approach to convert relational databases into column-oriented databases aims to solve part of these challenges, helping practitioners to perform migrations from traditional contexts to Big Data contexts.

For future work, one will study the concept of Big Data Warehouse (BDW), namely its main characteristics, changes from traditional environments, storage technologies, advancements in OLAP, query and integration mechanisms for BDWs, and implementations in specific contexts.

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Carlos Costa, MSc, is an invited lecturer and PhD student in Information Systems at the University of Minho. Previously, his academic formation was the Master’s in Engineering and Management of Information Systems at the University of Minho. His previous professional experience includes: web designer at QUATIC 2014; web developer and administrator at APSI/PTAIS; president at AIS/SC UMinho; software developer as a freelancer; database administrator and project manager at Conde Parish; and computer technician at Oversystems. His current R&D interests include: Data Science (data mining, data cleansing, data enrichment, and

data visualization); Big Data; Data Warehousing; OLAP; OLTP databases (SQL and NoSQL); software development; and Information Systems architectures. Mr. Carlos won the best paper award in the IAENG ICDMKE'15 conference.

Maribel Yasmina Santos, PhD, Habilitation, is an Associate Professor at the Department of Information Systems, University of Minho, Portugal. Maribel received the Aggregated title (Habilitation) in Information Systems and Technologies from the University of Minho in 2012 and a Ph.D. in Information Systems and Technologies from the University of Minho in 2001. She was the Secretary-General of the Association of Geographic Information Laboratories for Europe (AGILE) from May 2013 to June 2015 and Associate Director of the Department of Information Systems, University of Minho, from May 2010 to July 2014. Currently, she coordinates the Business and Location-enhanced Database Systems Research Track of SEMAG (Software Engineering and Management Group) at the ALGORITMI Research Centre, where she is a senior researcher. Her research interests include Business Intelligence and Analytics, Big Data Analytics, (Spatial) Data Warehousing and (Spatial) On-line Analytical Processing. Dr. Santos won the best paper award in the IAENG ICDMKE'15 conference.